SG No. 197 T^3 $I23$ [cubic]

* plus set: $+[0, 0, 0]$, $+\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$
* Wyckoff site: `2a`, site symmetry: `23.`

Table 1: Wyckoff bond: `6a@2a`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[X, 0, 0]$	[0, 0, 0]	`[1,-2,-3,4]`
2	$[0, X, 0]$	[0, 0, 0]	`[5,-6,-7,8]`
3	$[0, 0, X]$	[0, 0, 0]	`[9,-10,-11,12]`

Table 2: Wyckoff bond: `8b@2a`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	[0, 0, 0]	`[1,5,9]`
2	$[-X, -X, X]$	[0, 0, 0]	`[2,7,12]`
3	$[-X, X, -X]$	[0, 0, 0]	`[3,8,10]`
4	$[X, -X, -X]$	[0, 0, 0]	`[4,6,11]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	[0, 0, 0]	`[1,-2]`
2	$[-X, Y, 0]$	[0, 0, 0]	`[3,-4]`
3	$[0, X, Y]$	[0, 0, 0]	`[5,-6]`
4	$[0, -X, Y]$	[0, 0, 0]	`[7,-8]`
5	$[Y, 0, X]$	[0, 0, 0]	`[9,-10]`
6	$[Y, 0, -X]$	[0, 0, 0]	`[11,-12]`

Table 4: Wyckoff bond: `24d@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	[0, 0, 0]	`[1]`
2	$[-X, -Y, Z]$	[0, 0, 0]	`[2]`
3	$[-X, Y, -Z]$	[0, 0, 0]	`[3]`
4	$[X, -Y, -Z]$	[0, 0, 0]	`[4]`
5	$[Z, X, Y]$	[0, 0, 0]	`[5]`
6	$[Z, -X, -Y]$	[0, 0, 0]	`[6]`
7	$[-Z, -X, Y]$	[0, 0, 0]	`[7]`
8	$[-Z, X, -Y]$	[0, 0, 0]	`[8]`
9	$[Y, Z, X]$	[0, 0, 0]	`[9]`
10	$[-Y, Z, -X]$	[0, 0, 0]	`[10]`
11	$[Y, -Z, -X]$	[0, 0, 0]	`[11]`
12	$[-Y, -Z, X]$	[0, 0, 0]	`[12]`

* Wyckoff site: `6b`, site symmetry: `222..`

Table 5: Wyckoff bond: `6a@6b`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,-7,8]`
3	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,-11,12]`

Table 6: Wyckoff bond: `6b@6b`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4]`
2	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,-7,-8]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,-11,-12]`

Table 7: Wyckoff bond: `6c@6b`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,3,-4]`
2	$[0, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,7,-8]`
3	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,11,-12]`

Table 8: Wyckoff bond: `12d@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4]`
3	$[0, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6]`
4	$[0, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-8]`
5	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10]`
6	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-12]`

Table 9: Wyckoff bond: `12e@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[Y, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3]`
2	$[-Y, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4]`
3	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-7]`
4	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-8]`
5	$[0, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-11]`
6	$[0, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-12]`

Table 10: Wyckoff bond: `12f@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[1,-4]`
2	$[0, -X, Y]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,-3]`
3	$[Y, 0, X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,-8]`
4	$[Y, 0, -X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[6,-7]`
5	$[X, Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9,-12]`
6	$[-X, Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[10,-11]`

Table 11: Wyckoff bond: `24g@6b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[1]`
2	$[-X, -Y, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2]`
3	$[-X, Y, -Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[3]`
4	$[X, -Y, -Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[12]`

* Wyckoff site: `8c`, site symmetry: `.3.`

Table 12: Wyckoff bond: `8a@8c`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9]`
2	$[-X, -X, X]$	$[-x, -x, x]$	`[2,7,12]`
3	$[-X, X, -X]$	$[-x, x, -x]$	`[3,8,10]`
4	$[X, -X, -X]$	$[x, -x, -x]$	`[4,6,11]`

Table 13: Wyckoff bond: `24b@8c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, x]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -x]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -x, -x]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$[x, -x, -x]$	`[6]`
7	$[-Z, -X, Y]$	$[-x, -x, x]$	`[7]`
8	$[-Z, X, -Y]$	$[-x, x, -x]$	`[8]`

continued ...

Table 13

No.	vector	center	mapping
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$[-x, x, -x]$	[10]
11	$[Y, -Z, -X]$	$[x, -x, -x]$	[11]
12	$[-Y, -Z, X]$	$[-x, -x, x]$	[12]

* Wyckoff site: 12d, site symmetry: 2..

Table 14: Wyckoff bond: 12a@12d, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	[1,-4]
2	$[0, -X, Y]$	$[-x, 0, 0]$	[2,-3]
3	$[Y, 0, X]$	$[0, x, 0]$	[5,-8]
4	$[Y, 0, -X]$	$[0, -x, 0]$	[6,-7]
5	$[X, Y, 0]$	$[0, 0, x]$	[9,-12]
6	$[-X, Y, 0]$	$[0, 0, -x]$	[10,-11]

Table 15: Wyckoff bond: 12b@12d, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	[1,4]
2	$[-X, 0, 0]$	$[-x, 0, 0]$	[2,3]
3	$[0, X, 0]$	$[0, x, 0]$	[5,8]
4	$[0, -X, 0]$	$[0, -x, 0]$	[6,7]
5	$[0, 0, X]$	$[0, 0, x]$	[9,12]
6	$[0, 0, -X]$	$[0, 0, -x]$	[10,11]

Table 16: Wyckoff bond: 24c@12d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	[1]
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	[2]
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	[3]
4	$[X, -Y, -Z]$	$[x, 0, 0]$	[4]
5	$[Z, X, Y]$	$[0, x, 0]$	[5]
6	$[Z, -X, -Y]$	$[0, -x, 0]$	[6]
7	$[-Z, -X, Y]$	$[0, -x, 0]$	[7]
8	$[-Z, X, -Y]$	$[0, x, 0]$	[8]
9	$[Y, Z, X]$	$[0, 0, x]$	[9]
10	$[-Y, Z, -X]$	$[0, 0, -x]$	[10]
11	$[Y, -Z, -X]$	$[0, 0, -x]$	[11]
12	$[-Y, -Z, X]$	$[0, 0, x]$	[12]

* Wyckoff site: 12e, site symmetry: 2..

Table 17: Wyckoff bond: `12a@12e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,-8]`
4	$[Y,\ 0,\ -X]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,-7]`
5	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,-12]`
6	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,-11]`

Table 18: Wyckoff bond: `12b@12e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,4]`
2	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,3]`
3	$[0,\ X,\ 0]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,8]`
4	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,7]`
5	$[0,\ 0,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,12]`
6	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,11]`

Table 19: Wyckoff bond: `24c@12e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[12]`

* Wyckoff site: `24f`, site symmetry: `1`

Table 20: Wyckoff bond: `24a@24f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[-x,\ -y,\ z]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$[-x,\ y,\ -z]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$[x,\ -y,\ -z]$	`[4]`
5	$[Z,\ X,\ Y]$	$[z,\ x,\ y]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$[z,\ -x,\ -y]$	`[6]`

continued ...

Table 20

No.	vector	center	mapping
7	$[-Z, -X, Y]$	$[-z, -x, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, x, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, z, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -z, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -z, x]$	`[12]`